Integrating by parts in the second term we have

$$\begin{aligned}\beta^{-1}\int_{\Omega} f\Delta g\bar{\rho}_\beta \mathrm{d}\omega_q &= -\beta^{-1}\int_{\Omega}\nabla g^T\nabla(f\bar{\rho}_\beta)\mathrm{d}\omega_q \\ &= -\beta^{-1}\int_{\Omega}\nabla g^T[\nabla f\bar{\rho}_\beta - \beta\nabla U f\bar{\rho}_\beta]\mathrm{d}\omega_q \\ &= -\beta^{-1}\int_{\Omega}\nabla g^T\nabla f\bar{\rho}_\beta \mathrm{d}\omega_q + \int_{\Omega} f(\nabla g^T\nabla U)\bar{\rho}_\beta \mathrm{d}\omega_q.\end{aligned}$$

Combining this with the first part of the inner product, we see that

$$\langle f, \mathcal{L}g\rangle_\beta = -\beta^{-1}\int_{\Omega}\nabla g^T\nabla f\bar{\rho}_\beta \mathrm{d}\omega_q.$$

In this form we see that f and g appear symmetrically, so we may write

$$\langle f, \mathcal{L}g\rangle_\beta = \langle \mathcal{L}f, g\rangle_\beta,$$

hence, with respect to the weighted inner product, $\mathcal{L}^\dagger = \mathcal{L}$. (Note that the adjoint indicated here is quite different than the L^2 adjoint mentioned previously in connection with the Fokker-Planck operator.)

In addition to being self-adjoint, this operator (on the function space $C_0^2(\bar{\mu}_\beta)$) has other important properties. In particular, it has a *compact resolvent* [360]; this can be shown to imply that the spectrum consists only of a countable collection of eigenvalues, i.e. that the operator has a discrete spectrum. Moreover, from the calculation above,

$$\langle f, \mathcal{L}f\rangle = -\beta^{-1}\int_{\Omega}\|\nabla f\|^2\bar{\rho}_\beta \mathrm{d}\omega_q,$$

so that $\langle f, \mathcal{L}f\rangle \leq 0$ so the operator is negative definite and can only have non-positive eigenvalues. Many additional results are available for this case. A good summary of the state of the art is available in the forthcoming work of G. Pavliotis [298]. In particular, it is possible to show the properties of $\mathcal{L}$ contained in Theorem 6.1.

Theorem 6.1 *Define the generator of Brownian dynamics by*

$$\mathcal{L}f \stackrel{\text{def}}{=} -\boldsymbol{M}^{-1}\nabla U^T\nabla f + \beta^{-1}\Delta_M f.$$

Suppose that U is smooth with periodic boundary conditions $\mathcal{Q} = \mathbb{T}^{N_c}$ ($\mathbb{T}^{N_c}$ an N_c-dimensional torus). Then, with respect to the space $C_0^2(\mu_\beta)$ with inner product $\langle f, g\rangle_\beta$ as defined above, we have:

1. *$\mathcal{L}$ is self-adjoint, negative definite, with spectrum consisting of discrete eigenvalues on the half line $(-\infty, 0]$.*
2. *$\mathcal{L}$ has a* spectral gap, *meaning that, with exception of the zero eigenvalue, the remainder of the spectrum satisfies $\sigma(\mathcal{L}) \subset (-\infty, -\alpha]$ for some $\alpha > 0$.*
3. *averages with respect to the evolving, time-dependent distribution converge exponentially rapidly to averages with respect to the invariant distribution, and, for any smooth, compactly supported test function ϕ, we have*

$$|\bar{\phi}(t) - \mathrm{Av}_\beta(\phi)| \leq K\|\phi\|_\beta e^{-\alpha t},$$

for some positive K and α, where recall $\bar{\phi}(t)$ denotes the time average of ϕ defined in (5.6).

Thus Brownian Dynamics is ergodic for the configurational Gibbs-Boltzmann distribution.

6.4.4 Ergodicity of Langevin Dynamics

The key feature of Brownian dynamics that makes it possible to verify the ergodic property is the fact that each variable of the system is directly in contact with an independent stochastic Wiener process. This ensures that at each point of phase space, all possible directions are sampled and the paths will have freedom to move in any direction. In Langevin dynamics, taking $\boldsymbol{M} = \boldsymbol{I}$ for simplicity,

$$\begin{aligned} \mathrm{d}\boldsymbol{q} &= \boldsymbol{p}\mathrm{d}t, \\ \mathrm{d}\boldsymbol{p} &= -\nabla U(\boldsymbol{q})\mathrm{d}t - \gamma\boldsymbol{p}\mathrm{d}t + \sqrt{2\gamma \mathrm{k_B T}}\mathrm{d}\mathbf{W}, \end{aligned} \tag{6.47}$$

as well as in a number of other important SDEs arising in molecular dynamics, the picture is less transparent. Stochastic noise is only introduced in some parts of the system, but it is not immediately clear that the stochastic character will be transferred through the internal dynamics of the system into all possible directions.

In these more complicated examples, we are able to demonstrate ergodicity without finding an explicit solution (as in the Ornstein-Uhlenbeck example) or study of the dynamics' generator (as in Brownian dynamics), given some assumptions on the behavior of solutions. We state (without proof) a powerful theorem on the ergodicity of degenerate stochastic diffusions, whose proof is essentially contained in [257, Theorem 2.5] (see also [44, 160, 161, 253, 266]). We denote by $\mathcal{B}_\delta(\boldsymbol{x})$ the open ball in $\mathcal{D}$ centered on the point $\boldsymbol{x}$ of radius δ, while $\mathcal{B}(\mathcal{D})$ is the Borel σ-algebra on $\mathcal{D}$ (see Sect. 5.2.1).